



Solutions for Dynamic Difficulty Adjustment in digital games: A Systematic Literature Review

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ARTICLE INFO

Keywords:

Digital games
Dynamic difficulty adjustment
Methods
Frameworks
Models

ABSTRACT

Dynamic Difficulty Adjustment (DDA) is an important aspect of game design aimed at balancing the difficulty level to enhance player experience and prevent frustration or game abandonment. This paper presents a Systematic Literature Review (SLR) that focuses on the implementation of DDA techniques in digital games. The objective of this study was to identify and analyze DDA solutions (models, methods, and/or frameworks) for digital games in the ongoing research agenda. Of the 547 studies found in four bibliographic databases (ACM, IEEE, Scopus, and Web of Science), 34 were selected. The results revealed a diversity of approaches, mainly involving Artificial Intelligence techniques (50%) and heuristics/parameters manipulation (47%), among other possibilities. In addition to reaffirming the open research problems described in the literature, the need for further research on generalizable, flexible, and modularized approaches is highlighted, allowing the integration of various DDA strategies while minimizing the disadvantages of each strategy and ensuring good results. In this sense, the conducted review brings solution directions based on the exchange of knowledge from self-adaptive systems.

1. Introduction

Dynamic Difficulty Adjustment (DDA) is an essential aspect of game design that aims to automatically balance the difficulty level in digital games to avoid gameplay experiences that are either too easy or too challenging, which can lead to player frustration or game abandonment [1,2]. Adaptability in digital games has emerged as a significant research demand, and adjusting game parameters according to player experience is a promising solution [3]. DDA has been extensively discussed in the field of adaptability to digital games.

The central goal of DDA is to uphold equilibrium between the game's challenge level and the player's capabilities. When these two aspects are harmonized, the player enters a state known as the “flow channel” [1,2]. Being outside this zone leads to annoyance and impedes player engagement [1]. Thus, a game implementing DDA should be able to discern when the player is outside the flow channel, typically through evaluation components that assess performance, and then adjust difficulty-related variables using a suitable adjustment mechanism [2].

As a relatively recent research area [2], DDA poses several open questions and challenges. These include the creation of new DDA methods or the optimization of existing ones [4], the development of tools that make easier the implementation of DDA mechanisms [4,5],

and the design of versatile tools and models applicable to various game genres that are compatible with popular game engines [4,5]. Thus, the *solutions* for DDA encompass three important dimensions for us: frameworks, models, and methods. We incorporated these terms into our study based on their presence in the literature and our observation that they offered promising outcomes in our initial search efforts [6–8]. They are conceived as follows.

- **Frameworks:** A framework is a software component designed to be reused, which embodies a generic solution for a broad problem. It offers common functionalities that can be utilized by various application programs [9–11].
- **Models:** In general, a model is an abstract representation of a software system that captures different aspects of the system, such as its functionality, structure, behavior, and interactions with other systems [12,13].
- **Methods:** A method can be defined as a systematic and structured approach to solving problems or performing specific tasks. A method comprises a set of steps, techniques, and practices that are followed to achieve a specific goal [14,15].

The overall objective of this study is to conduct a Systematic Literature Review (SLR) to identify and analyze solutions for implementing

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<https://doi.org/10.1016/j.entcom.2025.101041>

Received 14 November 2023; Received in revised form 27 December 2024; Accepted 15 October 2025

Available online 18 October 2025

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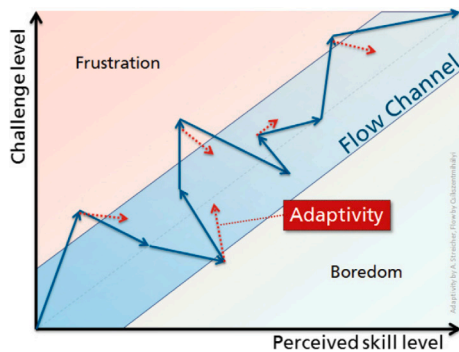


Fig. 1. The process of Dynamic Difficulty Adjustment [2].

Dynamic Difficulty Adjustment (DDA) techniques in digital games that have been the subjects of research in recent years. Our interest lies in comprehending the types of solutions under development, their characteristics and strategies, and the Dynamic Difficulty Adjustment (DDA) techniques that have been employed. We also sought to ascertain the purpose of the games involved in the research, whether for entertainment or serious purposes, and examine the evaluation processes employed to validate these solutions. Through an analysis of the identified works, our aim is to identify and address potential issues and existing gaps, thereby contributing to the understanding of the Dynamic Difficulty Adjustment field.

The following text is divided into five sections: Section 2 provides an overview of DDA concepts and the research area; Section 3 outlines the methodology employed in the review; Section 4 discusses aspects related to the execution of the review; Section 5 presents the results; and Section 6 concludes with final remarks and research opportunities.

2. Dynamic difficulty adjustment: Overview

Dynamic Difficulty Adjustment (DDA) is a fundamental concept in the field of adaptive gameplay in digital games. It involves automatic adjustment of game difficulty to maintain an optimal balance between the player's skills and the challenge level presented by the game. DDA aims to prevent players from becoming frustrated or disengaged in the face of extremely easy or difficult game experiences [1,2]. As the demand for adaptivity in digital games continues to grow, the use of DDA to adjust game parameters based on player experience has gained significant attention [3]. Although DDA research is relatively recent, it has emerged as a promising area of study in the field of adaptive gameplay in digital games [2].

The history of adaptivity and personalization can be traced back to the early days of the Internet, with the development of hypermedia systems that allow personalized content delivery. Over time, the concept of adaptivity has expanded to include adaptive e-learning systems, which involve personalized learning experiences based on individual knowledge and skills. More recently, these adaptive techniques have been used in digital games, with the introduction of artificial intelligence (AI) techniques playing a prominent role [1,2].

The primary objective of Dynamic Difficulty Adjustment (DDA) is to maintain a balance between the game challenge level and the player's skills. When these two elements are properly balanced, players enter a state of optimal engagement, known as the "flow channel" (Fig. 1) [1,2]. If the balance is disrupted, players may become frustrated or bored, leading to decreased retention and enjoyment [1]. A game with DDA capabilities must be able to recognize when a player is outside the flow channel, typically through performance evaluation

components,¹ and adjust the relevant difficulty parameters by using appropriate adjustment mechanisms (Fig. 2) [2]. Three crucial questions arise regarding the construction of a DDA mechanism: *what* to adapt, *when* to adapt, and *how* to adapt.

The literature presents several approaches to implementing DDA mechanisms in games, as classified by researchers such as Seyderhelm and Blackmore [1], and also Monteiro [16]. According to these authors, the approaches can be broadly categorized into: heuristic-based techniques, often referred to as "parameter tweaking", which involves adjusting the temporal factors, task complexity, speed, and challenge level; AI-based approaches, that utilize Reinforcement Learning algorithms, genetic algorithms, and other AI techniques. Some techniques combine elements from both categories, such as procedural generation focused on difficulty adjustment, direct player assistance systems, in-game feedback, and dynamic scripting.

Despite the progress made in the field of DDA, several open research problems remain to be addressed, as highlighted in the literature. One significant challenge is the exploration of various DDA strategies in combination [1,17], as well as the development of modularized approaches for the flexible organization of DDA mechanisms [2]. Additionally, there is a need for the creation of new DDA methods or the optimization of existing ones [4], along with the development of tools that facilitate the implementation of DDA mechanisms, making the process more efficient and accessible [4,5]. Moreover, versatile tools and models are required, which can be applied to different game genres and are compatible with popular game engines [4,5]. The need for tools in this sense arises, and SLR is justified in this regard.

In conclusion, Dynamic Difficulty Adjustment (DDA) is an important aspect of adaptive gameplay in digital games that aims to maintain the balance between challenge and player skill. DDA has been evolved as a solution to address the demand for adaptivity, and its implementation involves recognizing player performance, adjusting difficulty parameters, and avoiding excessive benefits and artificial interventions. The literature presents diverse approaches, ranging from heuristic-based techniques to AI-based methods, to enhance player engagement and retention. However, several research challenges remain, including the combination of DDA strategies, modularized design approaches, optimization of existing methods, and the development of versatile tools. Continued research and development in this field are essential for advancing the understanding and implementation of DDA in digital games.

3. Research methodology

To conduct this research, a Systematic Literature Review was carried out, following the PRISMA² guidelines as well as the approach proposed by Nakagawa et al. [18], which consists of three stages: (I) definition of the review protocol, including research questions, keywords, synonyms, search strategy, and inclusion and exclusion criteria for the studies; (II) review conducting, involving searches, selection of works, extraction, and analysis of data from the selected works; and (III) results and discussion, which presents not only the results but also answers to the research questions. The Parsifal³ platform was selected to aid in the management of the process.

This section presents the review protocol (research question, terms used in the search string, and the selection criteria). The remaining stages are described as follows.

¹ At this point, "performance" is broadly defined as the collection of data representing a player's current state and the relationship between their skills and the game's difficulty. These data encompass diverse categories including in-game data, vital signs, and cognitive metrics [2,6].

² <http://www.prisma-statement.org/>.

³ <https://parsifal.org/>.

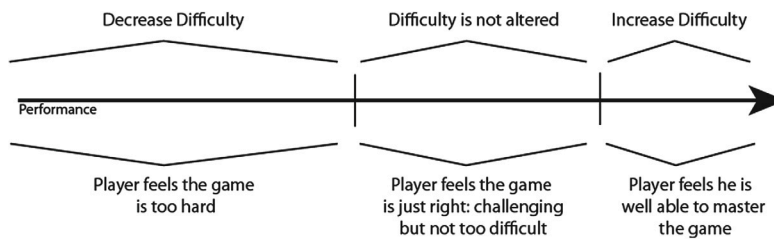


Fig. 2. Diagram illustrating the DDA mechanism's action, adapting difficulty according to performance [2].

3.1. Research question

Given the purpose of this research, one overarching research question have been formulated, based on previous studies [1,19]:

“What solutions are currently being researched for applying DDA in digital games?”

This is the primary focus of this review. It is important to understand the tools (specifically models, methods, and/or frameworks) that have been proposed in research, as the answer to that question has the potential to reveal paths that have been explored, as well as guide possible future directions of work. We have identified various factors intricately tied to this central issue, which primarily encompass the nature of the solution, the chosen strategies and approaches, the use of DDA techniques, along with the evaluation methods applied and the game's intent, i.e., entertainment or serious games. By considering these factors, the response to this question will not only provide us with a state-of-the-art perspective on the Dynamic Difficulty Adjustment field, but will also point out potential research opportunities and areas that still require further exploration.

3.2. Keywords, search string, and strategy

The following words were selected for the research given their high incidence in the scientific literature dedicated to the field: (P.1) *game (games)*, (P.2) *difficulty*, (P.3) *adjustment (adaptive, balancing, balance)*, and (P.4) *model, framework, or method*. These words were defined after several tests using control studies and related literature. The word *dynamic* was used initially, but its inclusion resulted in the loss of important studies. Therefore, they were discarded.

Thus, the search string was *(game*) AND (difficulty) AND (adjustment OR balance OR adaptive) AND (model OR framework OR method)*, considering the disjunctions between synonyms and conjunctions between keywords.

Regarding the strategy for conducting the review, we opted for a direct search on the selected bibliographic databases: ACM Digital Library,⁴ Elsevier Scopus,⁵ IEEE Digital Library,⁶ and Web of Science,⁷ as they are the main repositories related to the field of Computer Science and the development of digital games, and are used in related work [1,19].

3.3. Inclusion and exclusion criteria

According to Nakagawa et al. [18] and Sampaio and Mancini [20], to assess the relevance of the studies found, the application of inclusion and exclusion criteria is an essential process in an SLR. In this research, the criterion adopted for inclusion was “the work presents a model, method, and/or framework for the construction of a DDA mechanism in digital games”.

However, several criteria were adopted for study exclusion, and they were refined throughout the conduction process. The following criteria were used to exclude the studies:

- C1: Does not contain any of the keywords in the title and/or abstract
- C2: Not related to the context of digital games
- C3: Does not address Dynamic Difficulty Adjustment
- C4: Does not present a model, method, or framework
- C5: The full text of the work is not accessible
- C6: Not a scientific paper
- C7: Not written in English
- C8: Does not have an abstract or title
- C9: Work was published more than 5 years ago⁸

By applying these criteria to the studies identified using the developed search string, it was possible to determine which ones met the purposes of this research and were considered relevant to address the presented question.

4. Conducting

With the established planning, the *conduction* stage began, which involved conducting searches, selecting studies, extracting and analyzing the extracted data. Fig. 3 presents the flow of the search and study selection process.

The first step, therefore, was to conduct searches in which the search string was applied to the selected databases to obtain the results, which were cataloged on Parsifal. During this process, the search string was adapted (in terms of syntax) for each database, considering the specific requirements of each one. Exclusion criteria C6–C9 were directly applied to the databases to filter the results. A total of 547 studies were cataloged in the searches. After removing duplicate studies (192), the total number of studies was reduced to 355.

The selection process was accomplished. The studies found in the databases were evaluated based on the inclusion and exclusion criteria (C1–C5) used to classify the studies as *accepted* or *rejected*. In general, this was performed by reading paper titles, keywords, and abstracts. However, in some cases, the full text needed to be consulted. Finally, 44 studies were selected for full reading to extract and analyze the data.

During the selection process, the most frequently used exclusion criterion was C2, which resulted in the exclusion of 139 studies, corresponding to approximately 44.6% of the exclusions. This was followed by C3 (110 studies — 35.3%), C4 (57 studies — 18.3%), C1 (3 studies — 0.9%), and C5 (2 studies — 0.6%). The order of application of the criteria was C1 to C5. This means that most of the excluded studies had some of the keywords, but were either not related to Digital Games (many were about Game Theory, for example) or were in the field of games but did not address DDA. Another significant portion was about DDA but did not present a solution (model, method, or framework).

After the selection process, the selected papers were fully read, and the data were extracted. At this stage, ten studies were excluded, either because they did not address the theme or because they presented

⁴ <https://dl.acm.org/>.

⁵ <http://www.scopus.com>.

⁶ <http://ieeexplore.ieee.org>.

⁷ <http://www.isiknowledge.com>.

⁸ We considered the publishing date informed by the database from 2019 up to July 2023. The choice of this particular time range, a common practice in systematic literature reviews, was motivated by our aim of obtaining an updated perspective on the research domain (state-of-the-art).

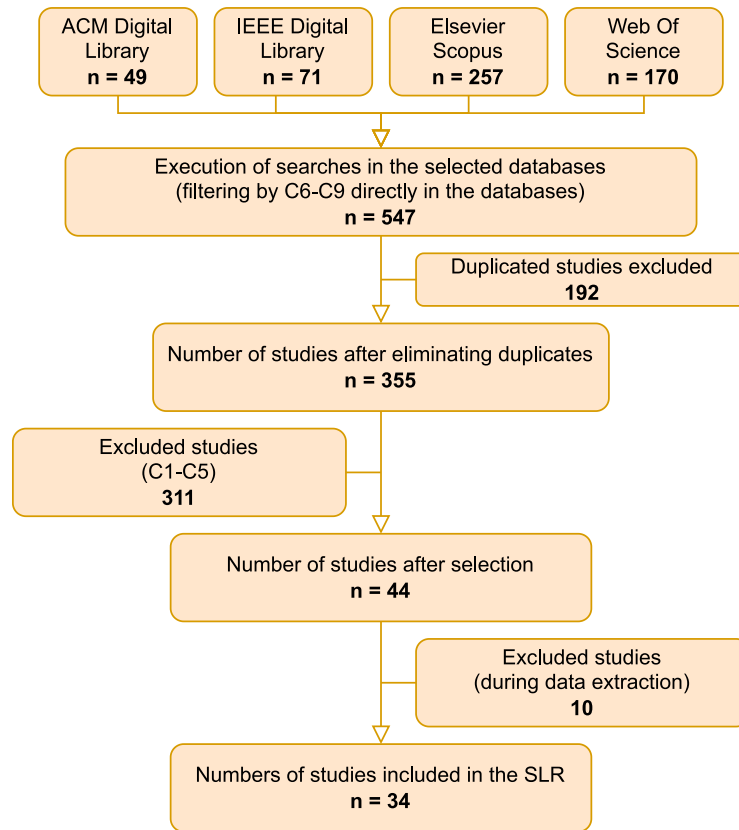


Fig. 3. Study search and selection flow.

Table 1
Accepted studies by database.

	Accepted	Citations
ACM	7	[21], [22], [23], [24], [25], [26], [27].
IEEE	5	[28], [29], [30], [31], [32].
Scopus	16	[7], [33], [34], [35], [8], [36], [37], [38], [39], [6], [40], [41], [42], [43], [44], [45].
Web Of Science	6	[46], [47], [48], [49], [50], [51].

similar studies by the same authors. Thus, 34 studies were included at the end of the selection process. The references and distribution of the studies by database are listed in Table 1. The reading of these studies, followed by analysis and discussion, allowed for the formulation of answers to the research question. The study selection and data extraction were reviewed by another researcher to minimize the risk of bias in the review. In addition, the extraction was performed using a structured form, also for this purpose.

5. Results and discussion

The studies selected for reading and data extraction presented various approaches and were able to provide an overview that allowed the construction of answers to the research question. Table 2 summarizes the reviewed DDA solutions, detailing their type (framework, model, or method), approach (AI, heuristics, or other), game's purpose (entertainment or serious), and evaluation processes (e.g., player experience or

proof of concept). The results presentation and discussion follows the structure provided in the table.

In terms of publication year, the studies were distributed according to the charts presented in Fig. 4. The first chart (Fig. 4a) disposes the 547 found studies according to the respective databases, and the second chart (Fig. 4b) plots the distribution of the 34 selected studies. The year with the highest number of papers accepted was 2019, with 11 papers, corresponding to approximately 32.3% of the selected studies, followed by the years 2020 and 2023, each one with 9 studies. Finally, the other studies were distributed between the years 2021 (2) and 2022 (3).

5.1. Solutions: Frameworks, methods, and models

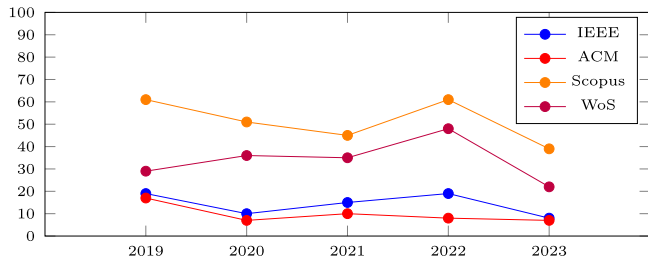
Several solutions have been presented in the selected studies. Regarding the type of solution (Fig. 5), there were 14 self-titled frameworks (41.2%), 12 methods (35.2%), and 8 models (23.6%).

The frameworks encompass different dimensions of DDA. Some frameworks aim to provide solutions for specific issues in DDA-based systems, ranging from player profiling classification [38] to assisting game designers in the implementation of specific features [48]. There are also works that address more specific aspects, such as the calibration of Reinforcement Learning algorithms used to build a solution [34] or the creation of behaviors with varying levels of difficulty for game agents [29].

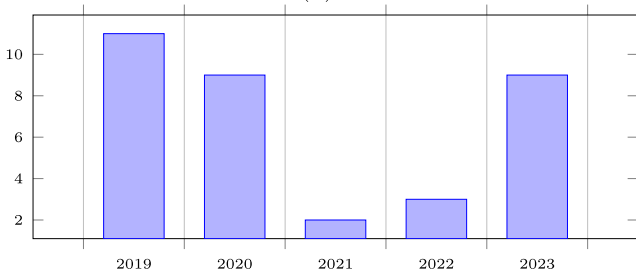
However, other generic studies aim to encompass the DDA mechanism as a whole or, in a broader sense, conceive the DDA mechanism as a cog in a larger mechanism, which is the case with projects such as BinG, which utilizes player provenance data to balance the difficulty [41], El-Habr's DDA mechanism coupled with procedural content generation [51], and the approach of Seyderhelm et al. [6], which proposes a modularized conceptual framework (Fig. 6) based on the Design–Implement–Evaluate triad. Those three great unities are composed by many minor interdependent modules. One module of the

Table 2
Accepted studies overview.

Studies	Solution type	Approach	Game's purpose	Evaluation
Afyouni et al., 2019 [21]	Framework	Heuristics	Serious	Other
Li et al., 2021 [22]	Framework	AI, Other	Entertainment	Other
Moon and Seo, 2020 [23]	Method	AI	Entertainment	Player Experience
Goutsu and Inamura, 2023 [24]	Method	AI	Entertainment	Player Experience
Schneider and Graham, 2023 [25]	Model	Heuristics	Entertainment	Player Experience
Sithungu and Ehlers, 2023 [26]	Method	AI	Entertainment	Proof of Concept
Zeng and Sweetser, 2023 [27]	Framework	Heuristics	Entertainment	Player Experience
Correa et al., 2022 [28]	Model	Heuristics, Other	Entertainment	Proof of Concept
Paduraru and Paduraru, 2019 [29]	Framework	AI	Entertainment	Other
Said et al., 2019 [30]	Method	Other	Serious	No Evaluation
Hendrix et al., 2019 [31]	Method	Unspecified	Entertainment	Player Experience
Lee et al., 2023 [32]	Method	AI	Serious	Other
Fujita, 2022 [7]	Method	AI, Other	Entertainment	Proof of Concept
Willwacher and Korn, 2021 [33]	Framework	Heuristics	Serious	No Evaluation
Cui et al., 2020 [34]	Framework	AI	Entertainment	Proof of Concept
Gonzalez-Duque et al., 2020 [35]	Method	AI, Heuristics, Other	Entertainment	Proof of Concept
Sandeep et al., 2020 [8]	Model	AI	Serious	Proof of Concept
Sarkar and Cooper, 2020 [36]	Model	Heuristics	Entertainment	Player Experience
Chanel and Lopes, 2020 [37]	Model	AI	Entertainment	Player Experience
Suryapranata et al., 2020 [38]	Framework	Unspecified	Entertainment	Player Experience
Pratama and Krisnadhi, 2019 [39]	Method	AI, Other	Entertainment	Proof of Concept
Seyderhelm et al., 2019 [6]	Framework	Heuristics	Serious	No Evaluation
Shakhova and Zagarskikh, 2019 [40]	Model	AI	Entertainment	No Evaluation
Figueira et al., 2019 [41]	Framework	Heuristics	Entertainment	Player Experience
Ajani and Mallipeddi, 2023 [42]	Framework	Heuristics	Serious	Player Experience
Chrysafiadi et al., 2023 [43]	Model	AI, Heuristics	Serious	Player Experience
Lyu and Bidarra, 2023 [44]	Method	Heuristics	Serious	Other
Shum et al., 2023 [45]	Framework	AI, Heuristics	Serious	Proof of Concept
Moon et al., 2022 [46]	Model	AI, Other	Entertainment	Player Experience
Reis et al., 2020 [47]	Framework	AI, Heuristics	Entertainment	Proof of Concept
Refai et al., 2020 [48]	Framework	Unspecified	Entertainment	Player Experience
Ozkul et al., 2019 [49]	Method	Other	Serious	Player Experience
Pan et al., 2019 [50]	Method	Heuristics	Serious	Other
El-Habr et al., 2019 [51]	Framework	Heuristics	Serious	Other



(a)



(b)

Fig. 4. Distribution of the studies according to publishing year: (a) for the total of 547 studies; (b) for the 34 selected studies.

framework is responsible for detecting the player's initial knowledge, motivations, and experiences. After the game, an evaluation for adjustments is conducted in another module; performance measurements and cognitive load are disposed in two modules, and another one adjusts the difficulty. By proposing a modularized solution, it is possible to delve into the specificities of each mechanism and combine different

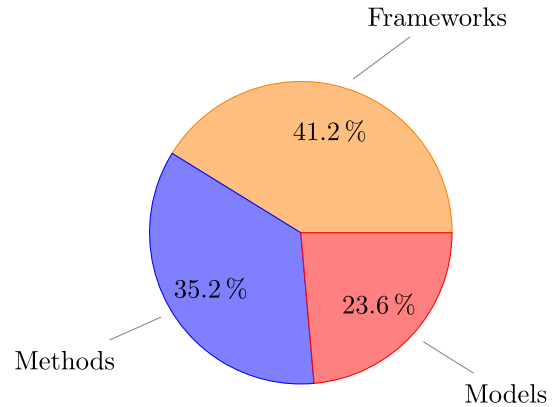


Fig. 5. Classification of the selected studies by solution type.

approaches. For the same purpose, the work by Shum et al. [45] aims to separate the DDA engine from the rest of the game's logic, and their framework is also composed of a series of modules.

On the other hand, approaches such as Zeng [27] are concerned with the DDA mechanism as a whole, but in a specific context related to games. It develops a framework for multiplayer games, in which it is necessary to balance the difficulty between players who are playing in the same room or environment. Based on the establishment of metrics and objectives, he focuses his approach on three major steps or definitions: collecting data, defining when to make the adjustment and, finally, how to do it.

A few studies have combined strategies. This is the case of Ajani and Mallipeddi, who combined heuristics with multi-objective optimization [42], and Rehabot [21], an adaptive gaming system designed

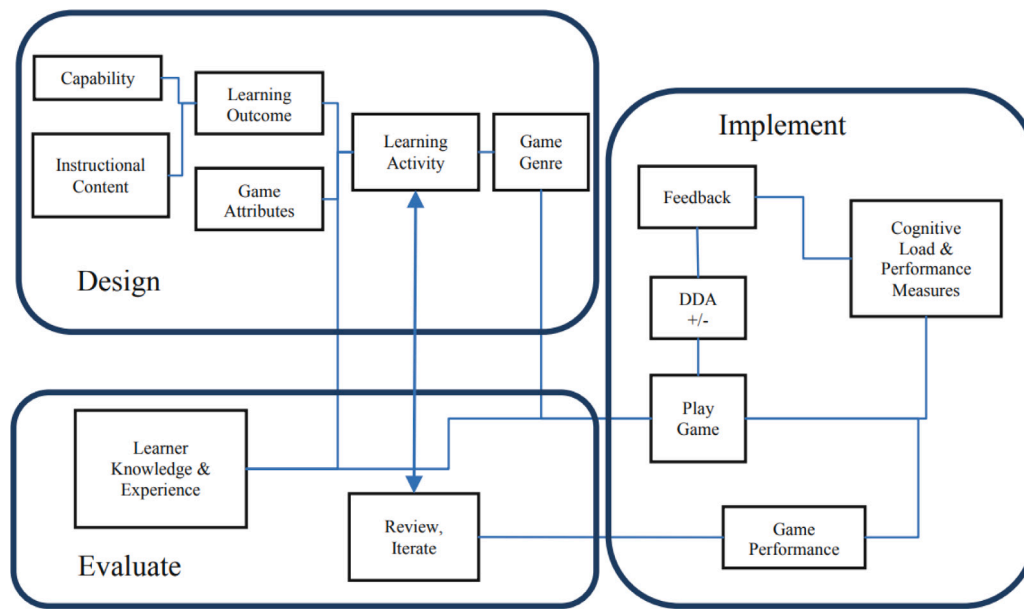


Fig. 6. Framework proposed by Seyderhelm et al. [6].

to guide patients in rehabilitation exercises, which utilizes difficulty-adjusted level generation and the presence of virtual assistants in game composition. Another approach was proposed by Li et al. [22], who presented a framework that combines player performance analysis with other categories of analysis (such as dynamic difficulty influence analysis) to prevent player churn.

Performance analysis is based on monitoring the challenge curve that the player traces throughout the game (Fig. 7). Assuming that challenges and skills must be balanced, i.e., in the “flow zone”, the greater the deviation from ideal values, the greater the probability of churn.

The framework (Fig. 8) is based on the Churn Prediction Classifier, which takes into account various inputs, including the player’s performance and difficulty (Difficulty Flow) model, comprehensive logs generated over their entire lifetime, and a general model that pertains to all users. This enables the classifier to conduct both general and specific analyses, allowing for comparative assessments among all users and specifically focusing on the player currently engaged in the activity. By considering both the player’s current performance and their historical data, the classifier generates forecasts.

Based on the predictions made by the classifier, the authors suggest implementing interventions within the system to proactively prevent churn. These interventions aim to address potential issues or concerns identified through the churn predictions, ultimately retaining the player’s engagement and participation.

Regarding the methods, a significant effort was made to apply concepts from other well-established knowledge areas to address challenges in the field of DDA. This is exemplified by the work of Fujita [7], who employed a deep neural network to adjust the difficulty of an AI opponent in zero-sum games (e.g., chess and Go). A neural network is utilized to classify the game difficulty based on the current state of the board, thereby adapting the skills of the AI opponent. The use of deep neural networks and Monte-Carlo Tree Search is also employed, similar to the approach proposed by Pratama and Krisnadhi [39]. Lee et al. combined DDA and machine learning [32]. Following a similar line of research, Moon and Seo proposed a DDA approach with rapid user adaptation based on deep neural networks, enabling a quick adjustment to a player’s abilities using a small amount of gameplay data [23]. Additionally, Gonzalez-Duque et al. introduced a method based on the application of Intelligent Trial-and-Error algorithm, which combines trial and error with machine learning to solve problems [35]. In

turn, Goutsu and Inamura used statistic approaches (Gaussian Process Dynamic Model) in order to measure the player performance [24].

Sithungu and Ehlers [26] propose an approach involving the development of a multi-agent classification system called AdaptiveSGA, which is designed to construct a DDA mechanism. The AdaptiveSGA system employs a combination of Reinforcement Learning, specifically utilizing the Q-learning algorithm, and Behavior Trees. The architecture of this system is built upon agents that embody the DDA key functionalities (as depicted in Fig. 9). These agents include the Perception Agent responsible for monitoring functions, the Classification Agent for analyzing the monitored data, a pool of Decision-Making Agents created by the Evolution Agent and selected based on the Classification Agent’s results, and finally, the Execution Agent that carries out system modifications. The agents share a common memory component known as the Shared Memory.

Some exceptions are the works of Pan et al. [50], which proposed the use of parameter manipulation, the approach by Ozkul et al. [49] based on obtaining player data through biofeedback sensors and the use of algorithms that adapt the difficulty based on these data, and the method proposed by Hendrix et al. [31], which outlines six steps for the construction of a DDA mechanism, systematizing, and synthesizing what is generally discussed in the current literature:

1. Determine relevant game variables that effectively indicate the player’s performance. In addition, establish suitable methods for quantifying these variables.
2. Identify game design elements that influence the game’s level of difficulty, which can be adjusted.
3. Specifically, pinpoint these variables within the actual implementation of the game.
4. Establish a connection between game mechanics and the identified variables.
5. Select an appropriate approach to utilize performance variables to calculate the difficulty of the game.
6. Establish initial values for the variables and consider the possibility of testing with players to fine-tune these values.

Along these lines, there is also the study by Lyu and Bidarra [44], who relate concepts within the adaptivity research field (procedural generation and player modeling). By the way, the study by Said et al. [30] deserves further attention, as it provides an examination

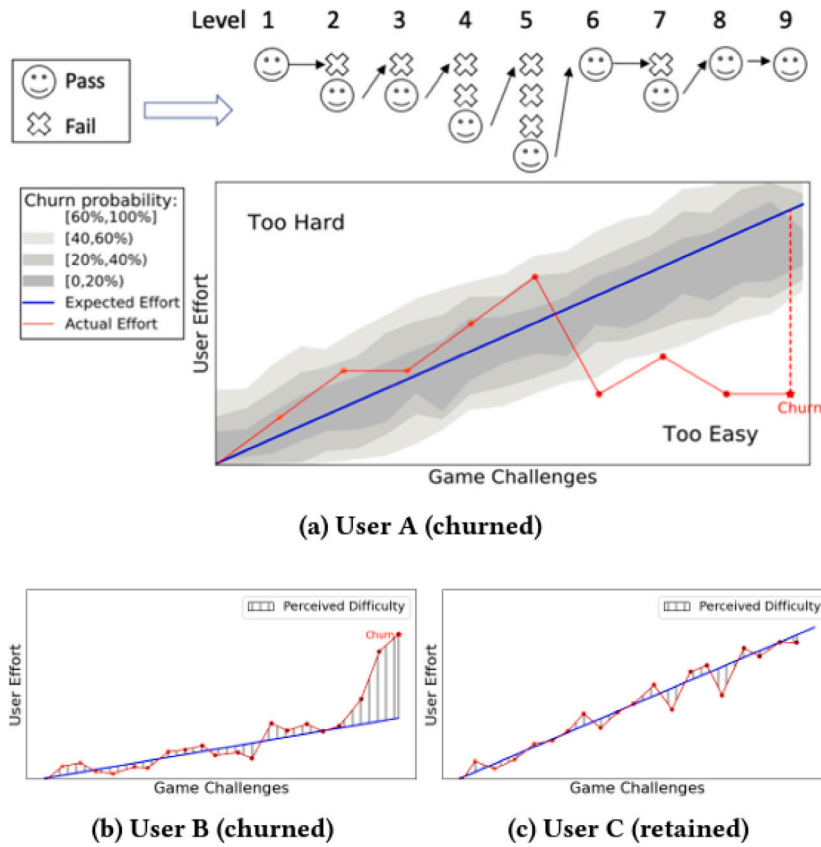


Fig. 7. Performance evaluation by Li et al. [22].

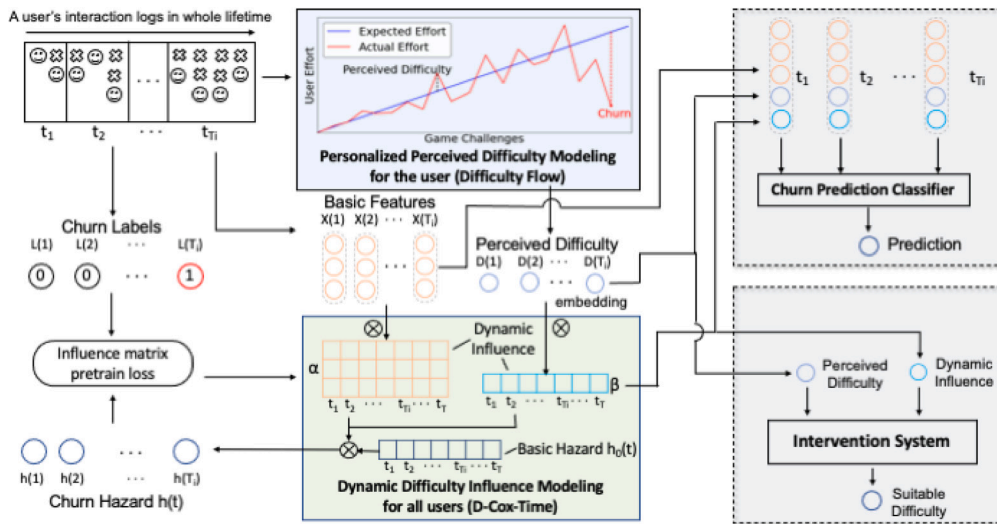


Fig. 8. Framework proposed by Li et al. [22].

of the variables to be considered concerning game design. Various interdependent concepts are presented in an ontology that transforms into a guide or a method for the conception of DDA mechanisms. These concepts are classified into *domains and competencies*, *game internals* (i.e., structure, context, rules, scenario, among others), and *player model*.

Eight studies were self-identified as models. Among these, four models are based on the application of machine learning techniques [8, 37,40,46]. On the other hand, Correa et al. introduces a model for generating the game's difficulty curve based on player activity. This

proposal is based on clustering performance characteristics of the players, evaluation functions with adjustments through policy corrections, population dynamics, and entity profile management [28]. In this case, the model is integrated into the game. Chrysafiadi et al. [43] purposes the use of fuzzy logic that receives a user model and a set of rules and adapts the game difficulty. In turn, Schneider and Graham [25] focus on aim assistance on shooting multiplayer games, presenting a heuristic model. Finally, Sakar and Cooper presented a heuristic-based method [36]. This study presents a model for combining two DDA techniques: rating systems (which seek to find a difficulty level according

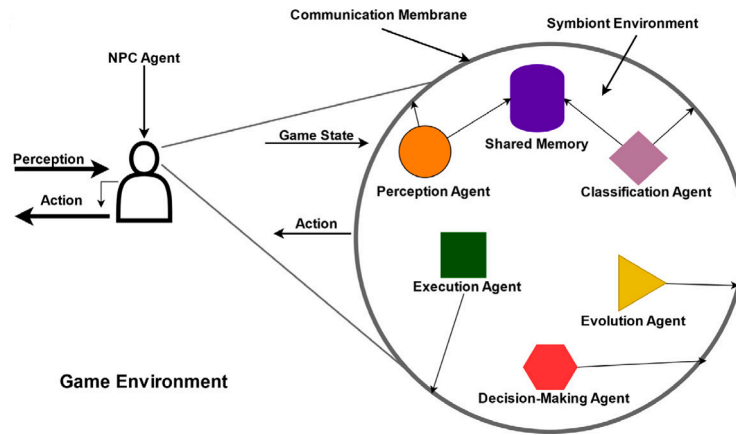
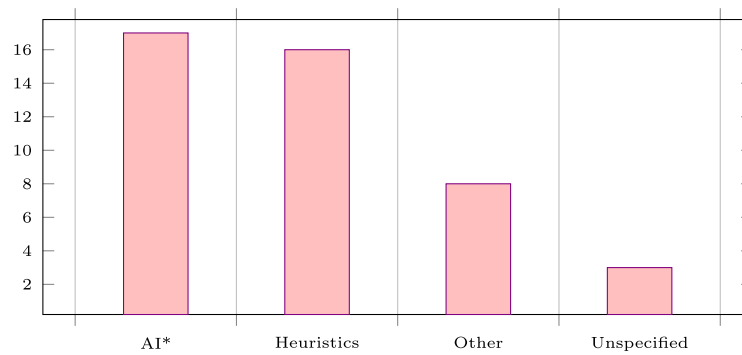


Fig. 9. Model proposed by Sithungu and Ehlers [26].



*AI includes mostly Reinforcement Learning, Machine Learning, Deep Learning and Neural Networks.

Fig. 10. Classification of the selected studies by used technique.

to the player's ability) and skill chains (that is, the progression of skills acquired by players throughout the game). Both concepts are related to in-game progression.

5.2. Strategies and approaches

In addition to the described solution type, an investigation was conducted on the techniques employed in each approach (Fig. 10). The majority of studies (17 studies) utilized Artificial Intelligence (AI) techniques such as Reinforcement Learning [47], in addition to Monte-Carlo Tree Search and Machine Learning (as previously mentioned), which were often used in combination [7,39,46]. Deep Learning [8,37] and Neural Networks [22,23] were part of many solutions.

For example, the framework proposed by Reis et al. utilized Reinforcement Learning [47]. This technique is widely cited in studies involving DDA because of its potential to eliminate the need for pre-training or a large amount of data at the beginning of the game. The approach in question involves incorporating a game master agent into the underlying game. The game master interacts with other players following a multi-agent systems (MAS) model and, constrained by a set of equilibrium restrictions, learns the optimal strategy to fulfill its equilibrium objectives.

Sixteen studies used heuristics, also known as parameter manipulation [6,25,28,36,42,44,50]. In general, these studies revolve around assessing a player's performance through intricate mathematical equations involving various game variables. By assigning specific weights to these elements and generating an evaluative value, these studies employ rules and heuristics to fine-tune the variables that determine the game's difficulty. This approach offers advantages that are cited by the studies, including versatility and seamless integration into

existing games. Moreover, it grants developers the ability to preemptively avoid undesirable events even before a possible first occurrence, further reducing player frustration. Another benefit is its low demand for extensive background data. However, a drawback lies in the considerable dependence on the game designer's prediction ability, which can complicate the implementation process [2,52,53].

Eight studies employed alternative approaches, either exclusively or in conjunction with the aforementioned approaches. Correa et al. combines heuristics with other mathematical and statistical models [28], Gonzalez-Duque et al. integrates AI techniques with the Intelligent Trial-and-Error algorithm [35], Said et al. focuses on constructing an ontology that enables the application of various approaches [30], and finally, Li et al. [22], as previously mentioned, presents a framework that combines multiple categories of analysis to prevent player churn. Three studies did not specify approaches but proposed solutions that can accommodate various approaches [31,38,48].

5.3. Game's purpose

Another factor we analyzed was the purpose of the game, that is, whether it was directed to pure entertainment or had serious purposes. Overall, 13 studies addressed solutions in the context of serious games, which corresponds to approximately 38.2% of the studies. Some studies merely mention the possibility of applying the proposed approach in the context of serious games; however, they limit themselves to that, without the concern of developing something specifically tailored to this category of games [30,43].

On the other hand, others delved more deeply. For instance, Willwacher et al. [33] presented a framework specifically focused on exercises involving patient rehabilitation, proposing an Artificial Intelligence-based adaptivity system aimed at generating personalized activities

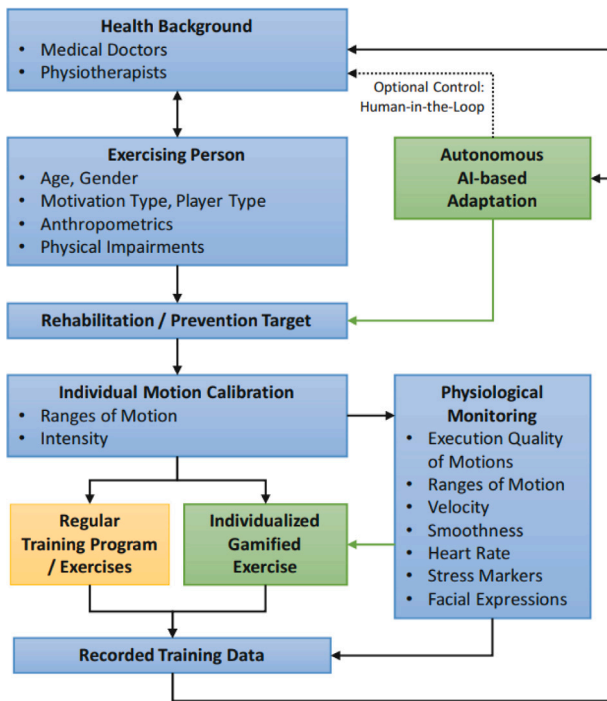


Fig. 11. Framework proposed by Willwacher et al. [33].

(Fig. 11). This is achieved by monitoring data such as exercise execution quality, speed, heart rate and facial expressions.

Based on a Health Background and information from the person who will perform the exercises (Exercising Person), a target is established and the Individual Motion Calibration module follows, which calibrates the movement ranges and their intensity. Physiological monitoring of data helps regulate exercise individualization. Activity data is recorded and used to improve the system.

Functional rehabilitation and exercise for health are very present in the selected studies. Ozkul et al. addressed upper limb rehabilitation [49], while Pan discussed a game based on free-movement training, where the patient can freely move to achieve predefined goals in the game's GUI [50]. Furthermore, Rehabot [21] introduced an adaptive gaming system with dynamic virtual assistants to guide patients during exercise. El-Habr's proposal [51] follows a similar line, focusing on motor activities for health promotion. Ajani and Mallipeddi [42] focus on arm rehabilitation, applying their framework in a two-player pong game, while Lee et al. [32] concentrate on trunk rehabilitation. In turn, Lyu and Bidarra [44] address motor rehabilitation, more broadly.

The cognitive and educational aspects were also addressed in the identified studies. Chrysafiadi et al. [43] such as Shum et al. [45] applied DDA concepts in games aimed at programming learning. Sandeep et al. investigated the use of Machine Learning techniques to estimate participants' cognitive abilities during working memory training, and explored whether they could provide more suitable training experiences by aligning training challenges more effectively with participants' needs [8]. On the other hand, Seyderhelm's study considers cognitive load by taking measurements using recognized methods employed in the literature [6].

Moreover, most of the studies that explicitly address serious games (the eight approaches discussed in the previous question) do not differentiate between the "serious" dimension of the game and the entertainment dimension. The general understanding, as evidenced by the selected sample, seems to be that there is no difference between these two aspects in the game design process of a serious game. The Rehabot project [21] works within this framework, without even differentiating the patient's health data, which are treated together with other game

data. Chrysafiadi et al. [43] advocates that the DDA mechanism must adjust the "game dimension", in a way that the player can focus only on the "serious dimension", separating DDA from the 'serious' context.

Some of these approaches deserve special attention. In the study conducted by Ajani and Mallipeddi [42], the essence of the suggested DDA solution originates from its established connection with functional rehabilitation context. Opting to employ multi-objective functions, the researchers cited one compelling reason: the necessity to effectively attain and harmonize conflicting objectives of intensity and performance.

In turn, Ozkul et al. used, in his approach to upper limb rehabilitation, physiological signals and performance values to assess players' levels of engagement [49]. Lee et al. [32] inserts patient's data (such as demographic data, patient's skills, and training sequence) in order to adjust the game's difficulty. Lyu and Bidarra [44] work exclusively with health data (therapy progression) to evaluate the player performance and to make the adjustments. However, Seyderhelm et al. points to a different direction: by incorporating the player's cognitive load data (measured data) into the project, the handling of these data is separate from the other data and is even treated separately on the performance calculation [6]. They suggested that some changes in game difficulty should be made considering only this cognitive load.

In summary, the selected works focusing on serious games are tailored to specific cases and are not easily generalizable to other situations. Therefore, there is a need for further research in the direction of generalizing the solutions and discussing the aspects in which serious games present specificities, specifically regarding their "serious" nature, that is, their application to a context that goes beyond entertainment. What does a DDA mechanism needs to fulfill to be applied in a serious game? Does developing a DDA mechanism for serious games simply involve applying something that is already directly applied to entertainment games? The discussion on the specificities of serious games was not sufficiently addressed in the studies of the sample in this review, which highlights the existence of research opportunities in this direction.

We also advocate that based on Seyderhelm's work, new research horizons involving DDA and serious games can emerge. It may be possible to advance towards providing specialized treatment for health variables and perhaps mechanics involving health data as a whole, involving a dedicated DDA mechanism for this purpose. The search for modularized approaches to game mechanics, as discussed in the previous questions, emerges as a possibility for more detailed work on the "serious" dimension of these games. Further research should be conducted in this regard.

5.4. Evaluation processes

The final category of analysis was the evaluation processes for the proposed solutions. In total, 30 studies (88.2%) addressed some form of evaluation process. In some of these studies, case studies were conducted, involving the application of the proposal in a real game and its utilization in time-defined matches, followed by the administration of questionnaires focused on player experience [23,24,27,36,37,41,42,46,49]. Other studies have discussed the applicability of the solutions in a specific game (proof of concept) or used simulated players (or similar) to analyze the performance evolution data [7,26,28,34,35,45]. Another forms of evaluation included cross validation (AI) [32] and also the assessment of concepts involving professionals in the game's application field, such as health care professionals [44].

5.5. Synthesis

In general, positive aspects were observed in the selected works of this review: the proposal of solutions that apply consistent concepts from various knowledge domains (within and beyond Computer Science and Digital Games) to the demands of Dynamic Difficulty

Adjustment (DDA), which enriches the approaches; the organization of the DDA mechanism as a separate system, facilitating the integration of the solution into target games; overall, these are approaches that can be applied to games of the same genre and/or scope with minimal modifications.

However, certain limitations of these studies can be highlighted in addition to the inherent limitations of each approach: many works are not comprehensive or extensible (i.e., they are heavily tied to a specific genre or game/context); in most works, there is limited discussion of the implementation process, including its difficulties and details, with the solutions being excessively high-level descriptions. Furthermore, the proposed DDA solutions do not take into account each game's mechanics and its specificities, but rather create a single DDA system that governs the entire game, assigning a single performance value to the player, while the diverse mechanics, each with their specificities, describe the game's difficulty in their own way [54]. We advocate that modularized approaches, that is, those that consider each mechanic in a specific manner and its contribution to the game's difficulty [55], can be good solutions to this limitation.

Moreover, given the observed trend towards the extensive use of Artificial Intelligence (AI) techniques in proposed solutions [26,39,46], the disadvantages of this usage are highly prominent. The need for prior training in many of the employed techniques, as well as the requirement for a large amount of data, significantly impacts the development of DDA mechanisms [5,47]. Furthermore, the inherently low explainability of AI solutions, that is, the limited ability to understand and explain how and why a particular algorithm makes its decisions or produces specific results, hinders solution enhancement. This becomes particularly crucial in cases involving serious games with health data, in which the decision-making process is of paramount importance [3,6,30].

Finally, we highlight that the discussed open problems in DDA were evidenced in this sample of papers. Few studies have explored the combination of multiple approaches (although these have shown success). The search for modularized and flexible approaches that can be applied in different contexts remains open. The search for tools to facilitate the implementation of DDA mechanisms has been extensively addressed within the scope of this SLR; however, the identified limitations provide avenues for further investigation in this direction.

In order to address these lacks, a promising solution should be organized, having as the "backbone" some well-known modularized approaches from other computer science areas. As digital games that work with DDA mechanisms can be considered self-adaptive systems⁹ [57], we can combine the established knowledge from that field with the specific requirements of adaptive gameplay.

Among the many techniques and tools available in the field, the MAPE-K loop is notable for its versatility, simplicity, and high level of modularization [56]. It is based on four activities or modules that form a cycle: Monitor, Analyzer, Planner, and Executor, which operate on a fifth component: the Knowledge Base. Therefore, it is possible to construct a solution that utilizes this MAPE-K cycle to consider the application of different techniques in each component, which can be harmonized within the loop. By adopting a highly modular approach, it becomes feasible to admit the creation of individual MAPE-K loops for each set of mechanics responsible for shaping a particular aspect of the game's difficulty. This opens the door to the seamless integration of multiple strategies within the game, potentially even within a single mechanics group.

⁹ A self-adaptive system is a computational system that can autonomously monitor, evaluate, and adjust its behavior and configuration to optimize its performance in response to changing environmental conditions or requirements [56].

6. Final remarks, limitations, and research opportunities

From this Systematic Literature Review, it can be concluded that many solutions have been subject of research for the construction of Dynamic Difficulty Adjustment mechanisms in Digital Games in recent years. The results showed a diversity of approaches, primarily involving techniques of Artificial Intelligence (69.2%) and heuristics and/or parameter manipulation (15.3%), among other possibilities.

However, some limitations are inherent to the process of conducting a Systematic Literature Review (SLR) [58]. The definition of the research question, bibliographic databases, and selection criteria can lead to the omission of studies and result in a selected sample that does not fully reflect the current state of research.

Furthermore, aside from these limitations, the present review was vulnerable to potential bias. The use of protocols such as ROBIS¹⁰ (Risk of Bias in Systematic Reviews) allowed for the identification of a risk of bias in the search process, because alternative methods were not used to search in bibliographic databases. In this case, relevant studies may not have been conducted because they were not indexed in the consulted bibliographic databases. However, the review of the steps by other researchers was a means to mitigate the risk of bias and discussed limitations.

Synthetically, the following research opportunities have been outlined: continuing to explore the research issues highlighted in the literature discussed in this study; dedicating efforts to uncovering generalizable, adaptable, and modular approaches that take into consideration the unique characteristics of each game mechanic and its impact on game difficulty, while also allowing for the integration of diverse DDA strategies to minimize their drawbacks and ensure favorable outcomes; in the realm of serious games research, conducting further investigations to generalize solutions and delve into the specific aspects that distinguish serious games, particularly concerning their non-entertainment applications; furthermore, there is potential to develop specialized frameworks for health-related variables and potentially incorporate health data-driven mechanics by implementing a dedicated DDA mechanism specifically designed for this purpose. The pursuit of modularized approaches concerning game mechanics can be instrumental in advancing these areas of inquiry.

CRedit authorship contribution statement

Carlos Henrique R. Souza: Methodology, Data curation, Writing – original draft, Investigation, Formal analysis, Writing – review & editing, Conceptualization. **Daniela F. Nascimento:** Investigation, Methodology, Data curation, Writing – review & editing, Writing – original draft. **Luciana O. Berretta:** Formal analysis, Writing – original draft, Conceptualization, Writing – review & editing. **Sergio T. Carvalho:** Formal analysis, Writing – review & editing, Conceptualization, Writing – original draft.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used OpenAI ChatGPT 3.5 in order to support translating the text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Funding

This work was supported by the Coordination for the Improvement of Higher Education Personnel (CAPES) [grant number 88887.683819/2022-00].

¹⁰ <https://www.bristol.ac.uk/population-health-sciences/projects/robis/>.

Declaration of competing interest

I hereby declare that there is no conflict of interest that could potentially compromise the integrity, objectivity, or impartiality of the research presented in this manuscript.

This research project is funded by a government scholarship from the Coordination for the Improvement of Higher Education Personnel (CAPES) [grant number 88887.683819/2022-00]. However, I affirm that CAPES or any other funding source had no involvement in any aspects of the study, including: study design; in the collection, analysis and interpretation of data; in the writing of the report; and in the decision to submit the article for publication.

On behalf of all authors, I affirm that there are no financial, personal, or professional interests that could be perceived as influencing the content of this research paper. This declaration is made in the spirit of transparency and in adherence to the ethical standards of academic and scientific research.

Data availability

No data was used for the research described in the article.

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